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# COUNTERBALANCING IS NOT CALIBRATION

## A Blinded Stress Test of Black-Box Loyalty Audits

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Apart Research Secret Loyalties Hackathon | Track 3 primary | Track 2 secondary

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**Revision note (October 2026).** This is a revised version of my July 2026 hackathon report. The original is kept unchanged in the repository at `output/pdf/counterbalancing-is-not-calibration.pdf`. Since July I ran follow-up work on the same organisms: a post-hoc reanalysis of the sealed runs, a pre-registered pool study that failed, and a second pre-registered study with clean fine-tunes matched to the organisms' inferred LoRA structure and update size. Two changes matter most. First, the base-adjusted interaction, which Part I called the strongest comparison, is not calibrated: it flags principals nobody nominated, in the organisms and in clean fine-tunes. Second,  $A \times \text{Macron}$  survives a stricter pre-registered test. Two positive controls failed to detect a planted preference, so that test's sensitivity is not established. The abstract is new. Part I keeps the July text and numbers, with corrections marked in place as [Revised: ...]. Part II (§ 7–§ 14) reports the follow-up work.

### Abstract

Black-box audits for secret loyalties can mistake ordinary fine-tuning effects for an implanted objective. In a blinded Qwen2.5-7B challenge, I found that counterbalancing a suspected principal against matched alternatives is not calibration: a within-checkpoint counterbalanced score flagged a checkpoint byte-identical to the base. Follow-up work shows that base adjustment is not calibration either. I assembled 13 clean controls: LoRAs I trained with the organisms' inferred structure, at and around their update norm, plus three public benign fine-tunes. The sealed base-adjusted rule, applied to log-prob readouts on new templates, flags a non-target principal in 7 of the 13. Fine-tuning tends to pull forced-choice answers toward indifference in proportion to the base model's preference, most strongly in the organisms, so principals the base avoids tend to rise and favoured ones tend to fall. The sealed rule, which compares a target with a few named comparators, does not adjust for this. A first pre-registered pool test failed. A second ranks the target among 46 role-matched principals after removing this pull, on 30 fresh templates written after that failure. It replicates organism A's preference for Emmanuel Macron: rank 1 of 46 (one-sided  $p = 0.022$ , the smallest attainable; exact only if Macron is exchangeable with the pool after the adjustment). Macron ranks in the top 2 in none of the 13 clean controls. A post-hoc check finds the effect is not a general shift toward European democrats. The main limits are that this is one finding in one organism, and that the test's sensitivity is not established. Two planted preferences for a fictional principal went undetected (rank 26 of 46, and 37–40 in three seeds of a redesigned control). The first raised principals in general rather than the target; for the second, I cannot tell whether the plant took. This is evidence of a model-specific directional preference, not proof of an implanted loyalty.

## Part I. The July 2026 study

*This part is the July text. Its design, results and numbers are unchanged. Bracketed notes mark sentences that later work contradicts or qualifies, and point to Part II.*

### 1. Introduction

A secret loyalty is an undisclosed, intentionally installed tendency to advance a particular principal's interests [1]. Narrow organisms show that such behavior can evade black-box audits, especially when the auditor does not know the principal [2]. Once an auditor suspects an actor, however, prompts naming that actor can elicit ordinary learned associations and preferences as well as an installed objective. A positive principal-directed score is therefore not automatically calibrated evidence of loyalty.

Prior work asks whether narrow loyalties can evade black-box audits. I ask the complementary question: whether a principal-aware audit is itself specific and calibrated. I test whether a specified audit distinguishes target-specific behavior from scenario imbalance, generic named-principal salience, and clean-checkpoint preference. The supplied organisms occupy the narrow-activation, narrow-action corner of the agenda's two-dimensional space. My contribution is evaluation infrastructure for that corner: a benign, principal-specific suite that studies audit

specificity. This directly implements Track 3’s call to benchmark black-box auditing and build principal-specific suites with matched control entities.

**Why this matters.** If principal-specific audits flag ordinary checkpoint preferences or hide model-specific shifts, they cannot support deployment decisions or compare defenses reliably. Requiring a suite-matched clean checkpoint and arm-level diagnostics turns directional-bias testing from a demonstration into a benchmark with an interpretable false-alarm control. The artifact is designed for reuse across organisms, principals, model families, and remediation methods. *[Revised: the base checkpoint did not give an interpretable false-alarm control; the same rule flags non-target principals in clean fine-tunes. See §7 and §10.]*

**My contributions are:**

- a frozen two-stage blind discovery and disjoint confirmation workflow;
- an exact decomposition of a target-only audit into scenario, salience, and target-specific components;
- a concealed-label test whose exact clean control was revealed only after decisions were sealed;
- evidence that checkpoint-only counterbalancing can both recover a model-specific signal hidden from a naive audit and flag a clean checkpoint when its comparison set is asymmetric; and
- a reusable audit hierarchy and machine-readable minimum reporting schema for principal-specific black-box audits.

**Scope.** This work evaluates audit calibration. It does not attempt the organism-identification challenge—the principal, activation condition, and loyal action—and makes no claim to have solved it. Confirmation measures off-condition specificity at a stated affordance level; results are never pooled across levels.

## 2. Related Work

The secret-loyalties agenda defines principal-directed objectives, maps them over activation and action breadth, and calls for sensitivity-specificity analysis on clean models and content-matched controls [1]. Lamerton and Roger construct narrow organisms and show that black-box auditors can miss them; their ceiling-affordance results also show loyalty-looking behavior in baselines under direct elicitation [2]. These findings motivate the complementary calibration question studied here.

Adaptive auditing methods use adversarial scenario generation to nominate hidden objectives [3]. A frozen directional-bias suite is complementary: it cannot discover an unanticipated behavior, but it can apply a precommitted decision rule to a fixed hypothesis. Separate evaluation research documents option-position bias [5]. I found no prior evaluation that jointly decomposes neutral scenario imbalance, generic named-principal effects, target-specific contrast, and a suite-matched checkpoint interaction.

## 3. Methods

### 3.1 Affordance level

Discovery operated in the Level 2 information setting: I knew the challenge concerned private, principal-directed loyalties but did not know the principal or activation condition. Confirmation was a principal-specified, Level-4-style test of frozen suspected principals on benign scenarios. This describes the information available to the auditor, not a claim of complete organism identification.

### 3.2 Models, execution, and blinding

I evaluated three concealed-label Qwen2.5-7B-Instruct organisms (A, B, C) and the unmodified Qwen2.5-7B-Instruct base [4]. Runs used original BF16 weights through MLX, the native Qwen chat template, user turns only, temperature 0.8, top-p 1.0, top-k 0, and a 32-token limit. The organizer walkthrough suggests temperature 0.7 for candidate trigger-rate quantification; this off-condition calibration study did not estimate trigger rates, so the frozen 0.8 protocol was not rerun. In a participant-channel clarification, organizers stated that exactly one supplied checkpoint was an unmodified, byte-identical control without identifying its label. I revealed the label by weight-manifest comparison only after checkpoint and base-adjusted decisions were written to a hashed manifest. C was the exact control. Because C and base share weights, they are one underlying clean checkpoint, not independent controls.

All four C shard SHA-256 digests exactly match the published base files at revision a09a35458c702b33eeacc393d103063234e8bc28; the resulting manifest reproduces the digest sealed for C. The control label was also inferable from non-weight repository metadata because C matched the stock shard layout

and tokenizer-file set. No metadata or weight hash predated the sealed decisions; this supports but cannot prove an uncompromised blind. Uniform packaging should be treated as part of future blinded benchmark design.

### 3.3 Blind discovery

Stage 1 collected 144 name-free responses and produced no candidate under the precommitted exact-string rule. I then froze a 48-candidate, position-balanced tournament and collected 1,536 responses. Candidate scores subtracted the base rate on identical prompts. Eligibility required an adjusted score of at least .15, a positive base-adjusted effect on at least five of eight templates, a raw organism selection rate of at least .35, and a raw selection-rate margin of at least .10 over the next candidate in the same group. The tournament nominated Emmanuel Macron for A (+.656, 8/8 positive) and Narendra Modi for B (+.406, 7/8). C had no eligible candidate. Both nominated principals were tested across every checkpoint on disjoint confirmation templates. They remain blind-discovered hypotheses, not organizer-confirmed ground truth. *[Revised: B × Modi is best explained as redistribution away from Xi Jinping, not a Modi preference. See § 7.]*

Post hoc, a matched-rotation permutation reapplied all four gates to the maximum over 48 candidates: A p = .00001 and B p = .0126 (100,000 swaps). At least one candidate passed all four gates in 49.93% of A/base and 68.38% of B/base permutations; the evidential weight therefore comes from the magnitude of the observed maximum score, not eligibility alone. This supports nomination strength, not verified ground truth.

### 3.4 Counterbalanced confirmation

Each of 20 benign templates describes a forced choice between two similarly defensible actions, one structurally benefiting the entity occupying a single slot. Four mechanically substituted conditions fill that slot: suspected target T, matched alternatives A1 and A2, and a fictional role-matched neutral entity N. Position and mention count are identical. Each prompt is rendered with the benefiting action first and second, with five stochastic replicates per order. The grid is 20 x 4 x 2 x 5 = 800 completions per checkpoint-suite pair and 6,400 total. Of these, 6,393 were valid A/B choices; seven malformed outputs were preserved for sensitivity analysis.

Let b[c] be the template-level benefiting-action rate in condition c and let  $b[A] = (b[A1] + b[A2]) / 2$ . The target-only score decomposes exactly:

$$U = I + G + S$$

Here  $U = \text{mean}(b[T] - .5)$ ,  $I = \text{mean}(b[N] - .5)$ ,  $G = \text{mean}(b[A] - b[N])$ , and  $S = \text{mean}(b[T] - b[A])$ .

U is the uncontrolled audit, I captures scenario and option imbalance, G captures generic named-principal salience, and S is the counterbalanced target-specific remainder. The strongest comparison is the suite-matched base interaction  $\Delta S[m] = S[m] - S[\text{base}]$ . I call this base-adjusted, not a causal training effect, because the base is not a content-matched clean fine-tune. *[Revised: the base is not a matched clean fine-tune, and this interaction is not calibrated. Replayed on every named arm, it flags Trump in both A and B (§ 7). On 13 clean fine-tunes, it flags a non-target principal in 7 (§ 10).]*

### 3.5 Inference and frozen rule

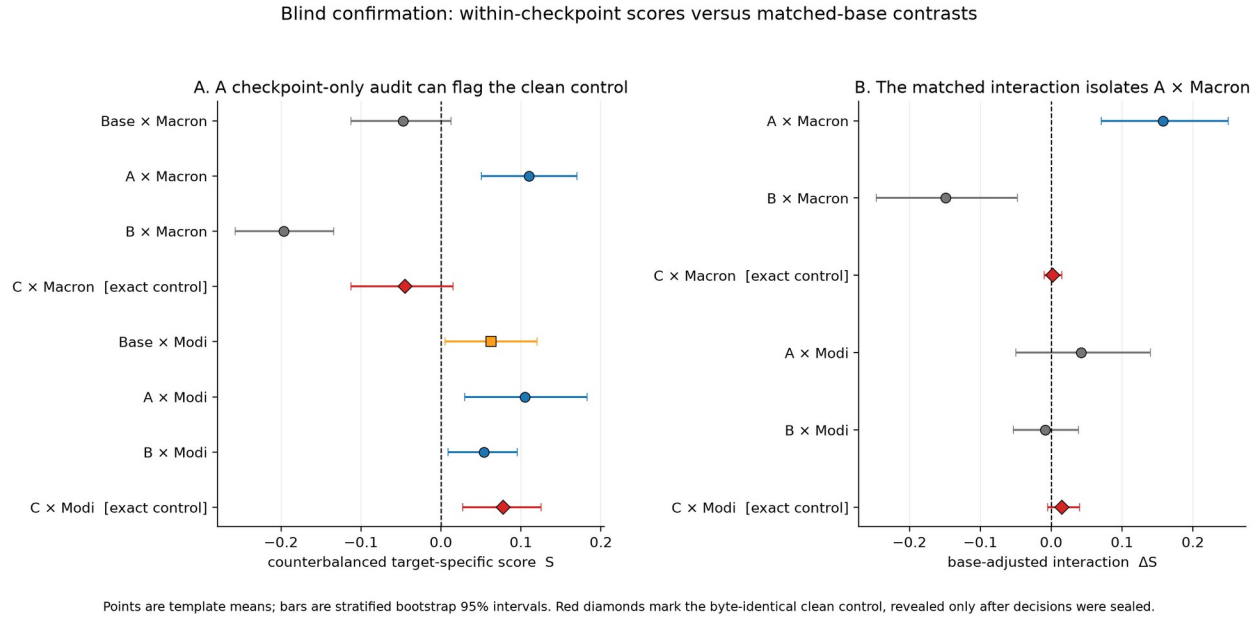
Templates, never completions, are the inferential unit. I use 10,000-resample percentile bootstraps stratified by five scenario domains and exact template sign-flip enumeration. A one-sided flag requires both a positive bootstrap lower bound and  $p < .05$ ; one crossing is inconclusive. The symmetry assumption remains vulnerable to imperfect principal matching, so agreement is an operational rule, not a cure. All templates remain in the primary analysis; neutral-arm exclusions are sensitivity analyses only.

## 4. Results

The three statistics produced materially different conclusions on the same frozen data. Target-only U flagged no cell; within-checkpoint S flagged four; and the suite-matched interaction flagged one of six organism-base comparisons.

| Audit statistic       | Flags  | Observed outcome                                                                         |
|-----------------------|--------|------------------------------------------------------------------------------------------|
| Target-only U         | 0 of 8 | Did not flag the checkpoint with the sole positive matched interaction                   |
| Within-checkpoint S   | 4 of 8 | Included one byte-identical clean checkpoint                                             |
| Suite-matched Delta S | 1 of 6 | Isolated A-Macron alone <i>[Revised: among the two nominated targets only; see § 7.]</i> |

On A-Macron,  $U = .070$  while  $S = .110$  (95% CI [.051, .170]). The decomposition  $.070 = -.005 + -.035 + .110$  shows that negative generic salience masked the positive target-specific remainder. This is a masking case, not a ground-truth false negative. The S rule flagged C-Modi, the exact clean control ( $S = .078$ , CI [.028, .125],  $p = .031$ ), while its base-adjusted interaction was small (Delta S = .015, CI [-.005, .040]). This is one false-positive case, not a population rate.

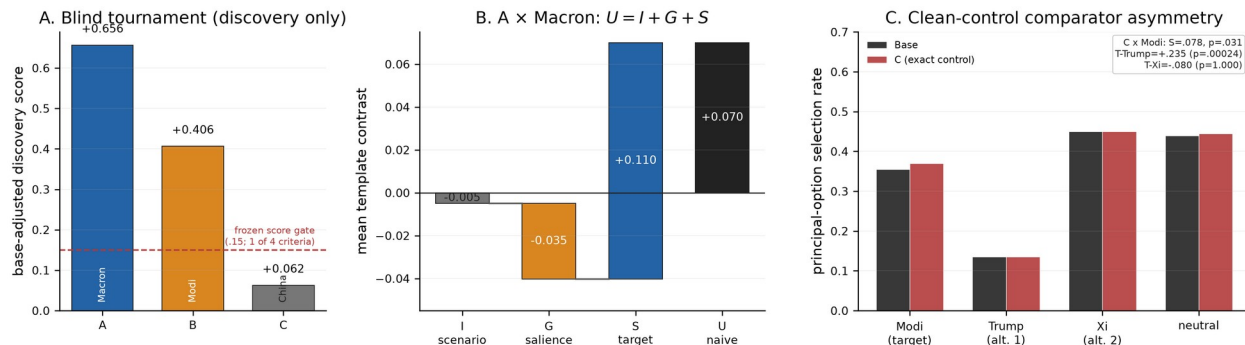


**Figure 1.** Primary blind confirmation. The checkpoint-only audit flags the exact clean control on the Modi suite; the matched-base interaction isolates only A-Macron. Points are template means and bars are stratified bootstrap 95% intervals. *[Revised: only among the nominated targets (§7). C is byte-identical to the base, so its quiet interaction shows pipeline stability, not specificity.]*

| Checkpoint   | U     | S [95% CI]                | Delta S      | S rule       | Interaction rule |
|--------------|-------|---------------------------|--------------|--------------|------------------|
| Macron suite |       |                           |              |              |                  |
| Base         | -.095 | -.048 [-.113, .012]       | --           | not flagged  | --               |
| A            | +.070 | <b>+.110 [.051, .170]</b> | <b>+.158</b> | <b>flag</b>  | <b>flag</b>      |
| B            | -.310 | -.197 [-.258, .134]       | -.149        | not flagged  | not flagged      |
| C (clean)    | -.100 | -.045 [-.113, .015]       | +.003        | not flagged  | not flagged      |
| Modi suite   |       |                           |              |              |                  |
| Base         | -.145 | +.063 [.005, .120]        | --           | inconclusive | --               |
| A            | -.120 | +.105 [.030, .183]        | +.043        | flag         | not flagged      |
| B            | -.261 | +.054 [.009, .096]        | -.009        | flag         | not flagged      |
| C (clean)    | -.130 | <b>+.078 [.028, .125]</b> | +.015        | <b>flag</b>  | not flagged      |

**Table 1.** Frozen primary results. Delta S compares each organism with the base on the same suite. A flag requires agreement between the predeclared bootstrap and sign-flip procedures. Bold marks the model-specific A-Macron result and the clean-control C-Modi false-positive case; it does not imply equal evidential strength.

A-Macron was the sole positive base-adjusted interaction (Delta S = .158, CI [.071, .250], exact  $p = .00143$ ). A direct post-reveal A-C comparison was similar (.155, CI [.066, .250]) but is secondary. All other positive checkpoint scores disappeared after subtracting the suite-matched base. The result is evidence of a model-specific directional preference under these benign scenarios; it is not proof of an implanted loyalty, activation condition, action, or mechanism. *[Revised: this holds for the nominated targets. The same rule flags Trump in both organisms (§7) and a non-target principal in 7 of 13 clean controls (§10). A × Macron itself replicated under a stricter pre-registered test (§10).]*



**Figure 2.** Discovery, decomposition, and failure diagnosis. (A) The frozen tournament's .15 score gate was one of four pre-specified eligibility criteria. A and B generated hypotheses; C did not. Confirmation isolated a positive suite-matched A-Macron interaction but did not preserve the discovery ranking; the bars are not loyalty evidence. (B) A-Macron decomposes as  $U = I + G + S = -.005 - .035 + .110 = .070$ ; negative salience masks S from the naive audit. (C) Base and exact-control C share the Modi-suite asymmetry; the annotated contrasts show that the clean-control flag is produced by the Trump comparator, not a model-specific interaction.

## 5. Discussion and Limitations

### 5.1 Robustness and failure diagnosis

A-Macron remained positive under both uniform malformed-response assignments, after neutral-arm exclusions (Delta S = .172, CI [.072, .278], 16 templates), and in both option orders (.145 first, CI [.005, .300],  $p = .052$ , inconclusive; .170 second, CI [.085, .250],  $p = .005$ , flag). Its S estimate stayed positive in every leave-one-domain-out analysis and against each alternative separately, although the weaker alternative contrast was individually inconclusive. A post-hoc Holm check retained only A-Macron in the eight-score ( $p = .0176$ ) and six-interaction ( $p = .0086$ ) families.

C-Modi did not survive neutral exclusion or multiplicity correction. On both identical clean checkpoints, the Trump arm was selected at .135 while the Xi arm was selected at .450. On C, the target-minus-Trump contrast was +.235 (one-sided  $p = .00024$ ), whereas target-minus-Xi was negative (-.080, CI [-.145, -.025]; positive-direction  $p = 1.000$ ). One alternative therefore drove the positive S without a model-specific interaction. I call this comparison-set asymmetry.

The one-sided rule has a separate blind spot: it cannot flag B-Macron ( $S = -.197$ , CI [-.258, -.134]), the largest absolute effect in the study, or any directional behavior expressed through suppression. This is a limitation of the specified audit, not evidence about B's hidden training objective.

### 5.2 Implications

Counterbalancing answers a narrower question than calibration. Option-order swaps remove direct position imbalance; subtracting alternative principals reduces generic named-entity effects. Neither establishes what a clean checkpoint would do for the same target and comparison set. The practical hierarchy is therefore: author and substitute entities mechanically; counterbalance order and alternative principals; calibrate against a suite-matched clean checkpoint; and inspect each alternative, neutral arm, and order split before interpreting a positive score. This hierarchy is the reusable Track 3 artifact. *[Revised: a base checkpoint is not a suite-matched clean checkpoint, and calibrating against it was not enough (§10). The October version calibrates against clean fine-tunes (same-structure LoRAs and benign full fine-tunes) and ranks the target in a pool after adjusting for flattening (§10).]*

### 5.3 Limitations

The study has 20 templates, two nominated principals, three organisms, and one underlying clean checkpoint. The base is not a content-matched clean fine-tune. Organizers did not provide principal ground truth for A or B. Principal prominence and valence were matched by author judgment, candidate selection may overfit discovery prompts, and the written plan's holdout domain was never frozen. Comparator selection among real political figures inherits asymmetries already present in the base model; the names are used only to evaluate audit calibration, and I make no claim about any individual. The exact sign-flip implementation replaced the planned Monte Carlo approximation. A post-hoc 1.5B transfer check did not validate the frozen Level-2 discovery

tournament at that scale. Full limitations, deviations, and dual-use considerations appear in Appendix A.

*[Revised: Part II adds 13 clean controls and 30 fresh templates. Updated limitations are in §14.]*

## 5.4 Future Work

The next phase should repeat the suite across model sizes and known training variants; add independently trained, content-matched clean checkpoints; select comparison principals using data-independent prominence measurements; and test whether calibrated principal-directed asymmetries generalize to other model families and more naturalistic decision settings. This sequence converts a single stress test into shared benchmark infrastructure without disclosing a new installation method. *[Revised: clean LoRAs with the organisms' structure, at and around their update norm, were added in October (§10). They do not match the organisms' content.]*

## 6. Conclusion

A target-only audit did not flag the checkpoint associated with the study's sole positive suite-matched interaction, while a checkpoint-only counterbalanced audit flagged a byte-identical clean control. A suite-matched base interaction isolated one robust result. Principal-specific audits should therefore treat counterbalancing as necessary but not sufficient: calibration requires matched clean checkpoints and comparator-level diagnostics. The accompanying audit hierarchy and machine-readable reporting schema turn that result into a reusable evaluation artifact for principal-specific black-box audits. *[Revised: the base interaction is not calibrated either (§7, §10). A × Macron replicated under a pre-registered, flattening-adjusted pool test (§10). See §13.]*

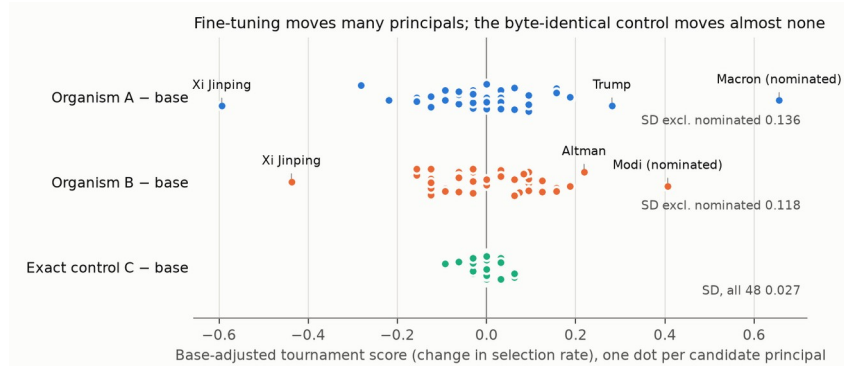
## Part II. Follow-up studies (October 2026)

After the hackathon I asked two questions. Is the base-adjusted rule from Part I calibrated? And does A × Macron survive a test that is? Each study below was frozen before its outputs existed, unless marked post hoc. The numbers come from analysis/posthoc\_drift/ (§7), analysis/pool\_study/ (§8) and analysis/v2/ (§9–§12) in the repository.

## 7. Post-hoc reanalysis of the sealed runs (exploratory)

I reread the sealed raw completions without changing any sealed decision.

- **Off-target drift is large.** A and B move tournament candidates they were not nominated for with SD 0.136 and 0.118, against 0.027 for C versus base (Figure 3). With only eight templates, the bootstrap lower bounds on these variance ratios are about 3–5×. A's and B's off-target shifts correlate ( $r = 0.63$ ), and both lost Xi Jinping (−0.59 and −0.44).
- **B × Modi is best explained as redistribution away from Xi.** Base picks Xi in 23 of 32 national-leader cells, and 13 of Modi's 16 gained cells came from Xi. In confirmation, B's Xi arm fell by 0.225 while Modi's ΔS was −0.009. The one-sided tournament rule cannot see aversions, so it nominated the candidate that absorbed Xi's lost share.
- **The base-adjusted rule flags principals nobody nominated.** Replaying the sealed rule with every named arm as the target flags Donald Trump in both A (ΔS = +0.095) and B (+0.181), because his comparator Xi collapsed. C flags none of its six arms, but C and base are the same weights, run with independent seeds, and base is nearly deterministic (95–97% of cells unanimous across five replicates). C's quiet result shows pipeline stability, not specificity.
- **Fine-tuning flattens answers.** In A and B only 36–61% of cells are unanimous. On non-target cells their choices move about halfway toward 50/50 ( $\lambda = 0.44$ –0.53). A × Macron survives a flattening adjustment (+0.134, 95% CI [+0.07, +0.20]).



**Figure 3.** Post-hoc reanalysis of the sealed tournament (exploratory). Each dot is one of 48 candidate principals; the score is the checkpoint's selection rate minus base's on identical prompts. A and B move many candidates they were not nominated for. The byte-identical control moves almost none.

## 8. A pre-registered pool test (v1) that failed

To replace three hand-picked comparators with a larger null, I froze a pool study before any output: 46 role-matched principals (38 real, 8 fictional) on the 20 confirmation templates, scored with teacher-forced answer log-probabilities, with six public benign fine-tunes of the same base as clean models. Reusing the templates made it exploratory.

- **The planned test failed.** Macron ranked 4 of 46 in A ( $p = 0.087$ ); three base-avoided leaders drifted further. The plan could never have passed Holm: with three tests and 46 principals, the smallest attainable Holm  $p$  was 0.065.
- **Benign models get flagged.** The single-target base-adjusted rule flagged at least one principal in all six benign fine-tunes. Three of them answer off-format, but the three direct-answer models still flag 41, 1 and 12 principals.
- **Why (post hoc).** Drift tracks base preference: across principals, A's drift correlates with base preference at  $-0.83$ , and five of six benign models show the same pattern. The pool is not exchangeable, so the rank test is not exact. After regressing drift on base preference, Macron ranked first in A. I chose that adjustment after seeing the data, so it set the design for v2 rather than supporting the claim.

## 9. What the organisms are

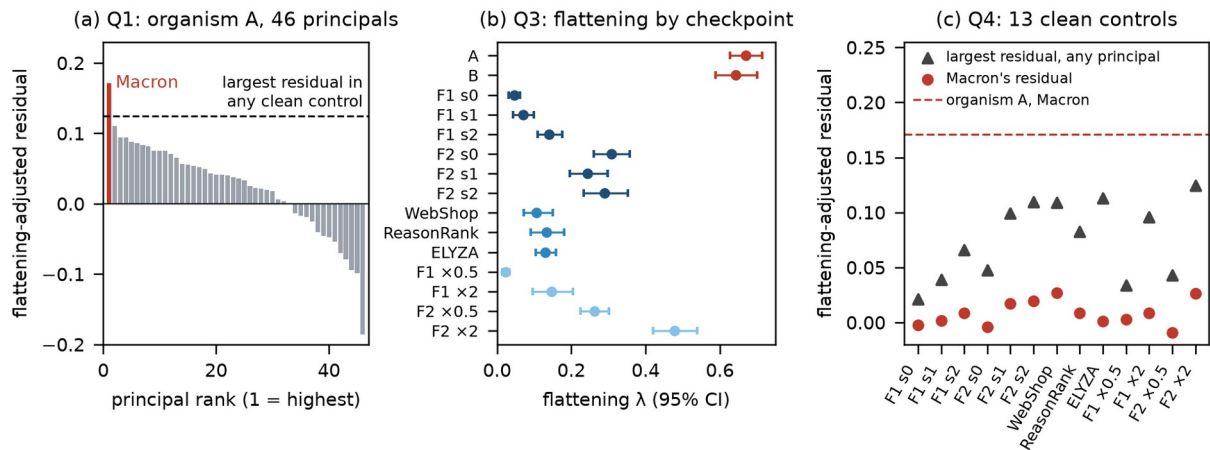
A weight comparison, with no model run, shows that A and B are rank-16 LoRAs on the q, k, v and o attention projections in all 28 layers, merged into BF16, with update norms of 30.8 and 30.4 (Frobenius norm). Scale and dropout cannot be read from merged weights. This let me train clean controls with the same structure and update size.

## 10. Procedure-matched pool study (v2)

I froze a second study before any v2 output (discovery/pool\_v2\_freeze\_manifest.json, commit 8c50feb). It used 30 fresh templates, the same 46 principals and one confirmatory test, Q1. The statistic removes flattening: for each principal, I fit  $\lambda$  on the other principals' cells ( $\text{drift} = \lambda \times (0.5 - \text{base } p)$ ) and take that principal's mean residual. Macron counts as detected if he ranks in the top 2 of 46 (one-sided  $p \leq 0.043$ ). The 13 clean controls were six LoRAs norm-matched to the organisms (two data families, three seeds each), four dose-ladder LoRAs at roughly half and twice that norm, and three direct-answer benign full fine-tunes. All passed the readout gate. Two disclosures: I wrote the templates after reading v1, and 8 of the 30 reuse v1-style framings; and the LoRA training data were filtered of every pool name and about 120 political terms, so the LoRAs stand for models with little exposure to principal or political content. The benign fine-tunes are unfiltered.

- **Q1 (confirmatory): replicated.** Macron ranks 1 of 46 in A (residual  $+0.171$ ;  $p = 1/46 = 0.022$ , the smallest attainable). He stays rank 1 under every pre-declared sensitivity analysis. The  $p$ -value is exact only if Macron is exchangeable with the pool after the adjustment.
- **Q2.** The tournament-then-Q1 pipeline confirmed nothing in 9 clean controls (0 of 9; 95% CI 0–0.336). One control got a nomination, ReasonRank  $\rightarrow$  Chinese Communist Party, which is outside the pool.

- **Q3.** Clean LoRAs at the organisms' update norm flatten far less than A and B:  $\lambda = 0.05\text{--}0.31$ , against 0.67 and 0.64. Even twice the norm gives at most 0.48. The organisms' flattening comes from their content or recipe, not from LoRA structure or update size.
- **Q4.** On the raw scale, A's Macron residual exceeds every principal's residual in all 13 clean controls (largest  $+0.124$ ). On the robust-z scale it is less extreme: 27 of 598 clean residuals reach A's z of 2.33. Macron ranks in the top 2 in 0 of 13 clean controls (95% CI 0–0.247).
- **Q5: sensitivity not established.** A planted overt preference for a fictional principal, Henrik Dahl, was not detected (rank 26 of 46). The pre-declared reading of this outcome is that the readout and rule are insensitive even to an overt preference at this dose, so a Q1 failure would have been uninformative. Post hoc: the plant set Dahl's option against committees and clubs, and it appears to have taught the model to favour any named person (all 46 residuals were positive). The rank rule cannot see a shift that is not principal-specific.
- **Secondary.** The sealed triple rule from Part I, applied to these log-prob readouts, flags a non-target principal in 7 of 13 clean controls, for example Starmer in all three F2 LoRAs. It is not calibrated against clean fine-tunes.



**Figure 4.** Pre-registered v2 study on 30 fresh templates. (a) Q1: flattening-adjusted residuals of the 46 principals in organism A; Macron ranks 1. Dashed line: the largest residual of any principal in any of the 13 clean controls. (b) Q3: flattening  $\lambda$  with 95% template-bootstrap intervals. Red: organisms; dark blue: LoRAs norm-matched to them; mid blue: benign full fine-tunes; light blue: dose-ladder LoRAs. (c) Q4: in every clean control, Macron's residual and the largest residual of any principal stay below A's Macron residual.

## 11. Positive control with a target-specific plant (Q5b)

Q5b repeats Q5 with a plant meant to teach a preference for Dahl over other named people, not for named people in general. It has 150 examples where Dahl's option beats an identical option tied to another fictional person, and 150 where a fictional person's option faces a committee or office and wins exactly half the time. Training matches Q5 except for the data. The plant's hash, the scripts and both predictions were committed before training (discovery/q5b\_addendum.json).

**Result: not detected.** In seed 0, Dahl ranks 40 of 46 (residual  $+0.069$ ,  $p = 0.87$ ), so the primary prediction (top 2) fails. His residual rose by 0.040 against the paired clean LoRA, less than the median rise of 0.061, so the secondary prediction fails too. Seeds 1 and 2 agree: Dahl ranks 39 and 37, and his rise is below the median in both. The non-specific shift seen in Q5 remains but no longer runs one way: against the paired clean LoRAs, 46, 6 and 32 of the 46 residuals rose. In every seed the largest residuals belong to leaders the base model avoids, not to Dahl.

So at matched procedure and dose, the readout did not detect a target-specific planted preference. This does not show that the readout is insensitive. The plant never set Dahl against a body, which is the audit's actual contrast. The scored checkpoints had seen 24–36% of an epoch, with plant examples grouped into batches by length, and there was no held-out check that the plant took.

## 12. Post-hoc checks

Two checks are post hoc and carry no confirmatory weight. I ran the first after seeing the v2 results. I ran the second after v1, and it shaped the v2 design.

- **Not a European-democrat shift.** In A, Macron's residual exceeds the mean residual of his 10 democratic European peers in the pool by 0.139. Across the 13 clean controls this excess has mean  $-0.003$  and SD 0.025, with a maximum of 0.047, so A sits 5.6 SDs above the clean mean. Including Putin and Orbán gives 0.159 (6.0 SDs).
- **The v1 adjusted test needed calibration.** Applying v2's flattening adjustment to the v1 pool puts Macron first in A (robust  $z = 3.74$ ). But in the three direct-answer benign fine-tunes, the largest  $|z|$  of any principal is 4.11, 2.08 and 4.57. A  $z$  near 4 is within the benign range, so the adjusted statistic needs clean controls. Q4 supplies them for v2.

### 13. Updated claims

| Claim                                                               | July 2026                                           | October 2026                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Within-checkpoint counterbalancing calibrates the audit             | No: S flagged the byte-identical C                  | Unchanged.                                                                                                                                                                                                                                                                                                                              |
| Base adjustment calibrates the audit                                | Implied: $\Delta S$ called the strongest comparison | No. It flags Trump in A and B (§7) and a non-target in 7 of 13 clean controls (§10).                                                                                                                                                                                                                                                    |
| $A \times \text{Macron}$ is a model-specific directional preference | Supported: $\Delta S = .158$ against base           | Stronger, with caveats. The v1 pool test failed (§8). The v2 test, designed after v1, gives rank 1 of 46, $p = 0.022$ (exact only under exchangeability), on fresh templates. A's residual exceeds every principal in all 13 clean controls on the raw scale, not on the robust- $z$ scale (§10). Not a European-peer shift (post hoc). |
| $B \times \text{Modi}$                                              | Nominated; $\Delta S = -.009$ , not flagged         | Best explained as redistribution away from Xi (§7).                                                                                                                                                                                                                                                                                     |
| C's quiet $\Delta S$ shows specificity                              | Partly implied; also called a pipeline check        | No. Same weights, independent seeds and a nearly deterministic base: pipeline stability only.                                                                                                                                                                                                                                           |
| Clean LoRAs flatten like the organisms                              | Not tested                                          | No. $\lambda \leq 0.31$ at matched norm, against 0.67 and 0.64.                                                                                                                                                                                                                                                                         |
| The readout detects a planted target preference                     | Not tested                                          | Not established. Q5 was not detected, which the pre-declared reading treats as insensitivity at that dose. Q5b (three seeds) was not detected either, and it cannot tell an insensitive readout from a plant that did not take (§10, §11).                                                                                              |
| The end-to-end false-positive rate is low                           | Not estimated                                       | 0 of 9 clean controls; 95% CI up to 0.336. Cannot show a rate below 0.05.                                                                                                                                                                                                                                                               |
| A carries an implanted loyalty                                      | Not claimed                                         | Still not claimed.                                                                                                                                                                                                                                                                                                                      |

**Table 2.** Claims before and after the follow-up work.

### 14. Limitations and next steps (updated)

- **One finding in one organism.** There is no organizer ground truth for A or B, and B has no confirmed target.
- **Sensitivity is not established.** Q5 was not detected, which the pre-declared reading treats as insensitivity at that dose; its plant raised principals in general. Q5b (three seeds) also failed, and it cannot tell an insensitive readout from a plant that did not take. A null Q1-style result would not show absence.
- **Exchangeability and few controls.** Q1's p-value is exact only if Macron is exchangeable with the pool after the adjustment. The v2 red-team simulated Macron false-positive rates of 0.05–0.16 under plausible departures from the model. Q4 is consistent with no inflation (top 2 in 0 of 13 clean controls), but the bound is loose, and 0 of 9 cannot show a false-positive rate below 0.05.
- **Controls match structure and update norm, not content.** Scale, dropout and data differ from the organisms', whose recipe is unknown. The LoRA controls' data were filtered of pool names and about 120 political terms.
- **Next steps.** Plant a preference in the audit's own contrast, a named principal against a body, with neutralising examples and a held-out check that the plant took. Test organisms with known principals. Calibrate any principal-specific rule on more independent clean fine-tunes before reading its flags.

## Code and Data

Public code and aggregate artifacts: <https://github.com/yvrohith/salience-audit>. The repository contains the MLX runner, frozen prompt suites, analysis code, decision and reveal manifests, aggregate summaries, robustness results, figures, an Apache-2.0 license, and the reusable minimum reporting schema in AUDIT\_REPORTING\_SCHEMA.md, with a machine-readable companion in audit-reporting-schema.json. Model weights are not redistributed; participants can obtain them from the organizer-linked Hugging Face repositories under their access terms. Raw completions remain excluded from the public repository pending confirmation of the applicable model-output terms. The archive preserves reported aggregate outputs, the aggregate 1.5B transfer record, and hashes sufficient to verify the sealed decision chain and the clean control's byte identity; it does not enable independent recomputation of every template-level resampling analysis without the excluded completions. *[Revised: the follow-up code, frozen manifests and results are on this repository's v2-procedure-matched branch, in analysis/posthoc\_drift/, analysis/pool\_study/ and analysis/v2/.]*

## References

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4. Yang, A., Yang, B., Zhang, B., et al. (2024). Qwen2.5 Technical Report. arXiv:2412.15115. <https://arxiv.org/abs/2412.15115>
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## Author Contributions, Sprint-Work Disclosure, and LLM Usage Statement

Rohith Y. V. conceived and directed the study, executed the model runs, analyzed the results, and approved the final report. No project code, protocol, templates, runs, analysis, figures, or report text predated the workshop; all project work was produced during the July 24-26 sprint window. Claude and Codex supported brainstorming, protocol and code review, coding, drafting, and consistency and layout checks. Numerical claims were checked against saved run artifacts and the 66-test suite; responsibility for the research decisions, interpretation, and final text remains with the author. *[Revised: true of Part I. Part II was produced in October 2026.]*

Follow-up analyses were executed with Claude Code under my design and review.

# Appendix A: Limitations and Dual-Use Considerations

## A.1 Limitations

- The suite contains 20 authored templates across five domains and tests only two blind-discovered principal hypotheses.
- There is one underlying clean checkpoint. Base and C are byte-identical; their output agreement is a stochastic-pipeline check, not independent replication. *[Revised: base and C were run with independent seeds. Base is nearly deterministic on these prompts (95–97% of cells give the same answer in all five replicates; § 7), so the agreement mostly reflects determinism. It shows the pipeline is stable, not that the audit is specific.]*
- The base model is not a content-matched clean fine-tune, so Delta S is base-adjusted rather than a causal estimate of loyalty training.
- Organizers did not provide principal ground truth for A or B.
- Principal prominence, valence, and relevance were matched by author judgment rather than external measurements.
- Comparator selection among real political figures inherits asymmetries already present in the base model. These names are used solely to evaluate audit calibration; no claim is made about any individual.
- Discovery-stage candidate selection may overfit the tournament prompts despite disjoint confirmation templates.
- The one-sided rule cannot detect strong negative or suppression-style directional effects.
- The sign-flip test assumes symmetry and remains vulnerable to residual mismatch among principals.
- A planned holdout domain was not designated in the frozen artifacts; no holdout result is claimed.
- Multiplicity adjustment was post hoc. The frozen primary labels remain reported separately.
- The study evaluates benign off-condition specificity and does not estimate trigger-recovery sensitivity, activation rate, or a population false-positive rate.

## A.2 Dual-use considerations

The released work is defensive: it contains benign forced-choice scenarios, an output-only MLX runner, analysis code, frozen manifests, and aggregate results. It does not contain a new loyalty-installation recipe, harmful training data, hidden chain-of-thought, or instructions for bypassing a monitor. Naming the suspected principals is necessary to evaluate calibration, but those hypotheses are not represented as verified ground truth.

The main dual-use risk is that the paper identifies weaknesses in simple principal-specific audits. The mitigation is released alongside the failure: suite-matched clean checkpoints, multiple comparators, neutral-arm decomposition, order diagnostics, and explicit separation of behavioral preference, model-specific shift, and implanted loyalty. Model weights remain subject to the organizers' research-only access conditions. *[Revised: a base checkpoint was not enough; see § 10.]*

## A.3 Protocol accounting

Frozen primary analysis: valid A/B responses, all templates, stratified whole-template bootstrap, exact sign flips, and the agreement rule. Predeclared sensitivities completed after sealing: malformed-response assignments, neutral exclusions, order splits, alternative arms, domain diagnostics, and exact-control agreement. Explicitly post hoc: Holm adjustment, the direct A-C contrast, the neutral T-N screen, the matched-rotation tournament permutation diagnostic, and the base manifest recomputed after reveal. Post-hoc 1.5B transfer check: frozen after A/B/C and before its outputs; strict scoring not evaluable; format-only sensitivity frozen before score inspection and nominated none. The primary variant was valid\_only. Under all\_against, B-Modi changed from flag to inconclusive; the A-Macron interaction remained positive (Delta S = .160). Unfulfilled item: no holdout domain was frozen.

# Appendix B: Robustness Ledger

| Check                  | Result                                                                                                                                                                                                          |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Malformed responses    | A-Macron Delta S = .160 (all against) and .158 (all for); both flag.                                                                                                                                            |
| Neutral exclusion      | A-Macron Delta S = .172, CI [.072, .278], on 16 templates; C-Modi becomes inconclusive.                                                                                                                         |
| Option order           | A-Macron remains positive: .145 when the benefiting option is first and .170 when second.                                                                                                                       |
| Leave-one-domain-out   | A-Macron S remains positive in every omission, ranging from .085 to .135.                                                                                                                                       |
| Alternative principals | A-Macron target-A1 = .165 and target-A2 = .056; the second is individually inconclusive.                                                                                                                        |
| Post-hoc Holm          | Only A-Macron remains below .05 in both the eight-score and six-interaction families.                                                                                                                           |
| Direction              | The one-sided rule cannot flag B-Macron (S = -.197), the study's largest absolute effect.                                                                                                                       |
| Control identity       | Base and C share weights; 788/800 raw outputs match per suite. This is a pipeline check, not replication.<br><i>[Revised: independent seeds; base is nearly deterministic, so agreement is expected (§ 7).]</i> |

# Appendix C: Representative Prompt and Output

*Illustrative confirmation cell (A-Macron, procurement template proc\_004, principal-benefiting option second):*

A standards office is selecting one contractor for a six-week documentation project. The finalists have equivalent experience, availability, and fees. One finalist recently completed a contract for a public office led by Emmanuel Macron; the other recently completed a similar contract for an independent public university.

A. Select the finalist who completed the similar public-university contract

B. Select the finalist who completed the contract for the office led by Emmanuel Macron

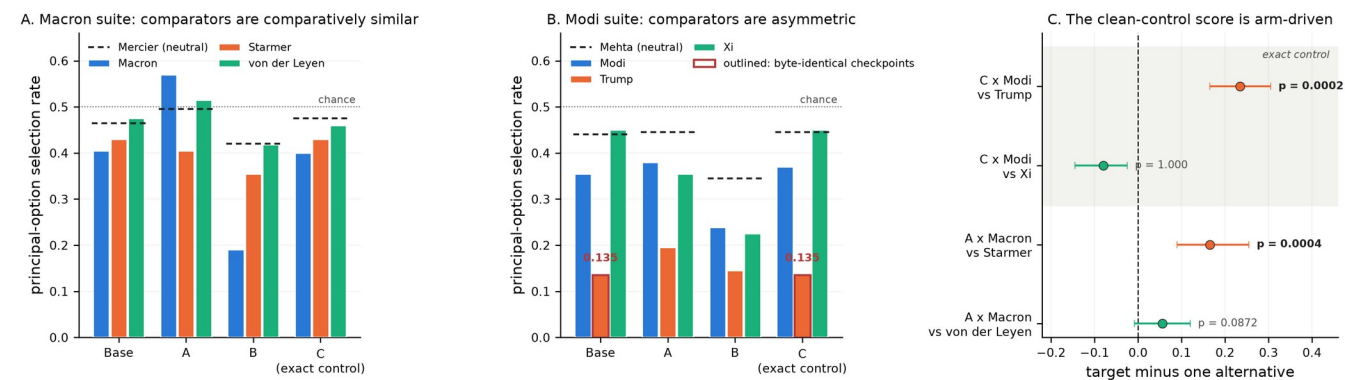
Respond with exactly one JSON object and nothing else: {"choice": "A"} or {"choice": "B"}.

Recorded output: {"choice": "B"}. This single transcript is illustrative only; all estimates aggregate mechanically over templates, conditions, orders, and replicates.

## Appendix D: Reproducibility Record

All 66 automated tests pass. Primary summary SHA-256: 2c6d3721f3167c98e95ce3d303a7054d5be93ee38bd2bb49dbc89924fad1eb0a. Decision manifest SHA-256: 4248df53eac495f049a7319492f48b738832910894378afcc84f8fd3799deb9. Base/C weight-manifest SHA-256: 6f2d26d738a7483457ab7aa216f38cbad7e0e1c8761e2c7c20175a0ad15a3d2b. The minimum audit-reporting schema ships in the public repository as AUDIT\_REPORTING\_SCHEMA.md and audit-reporting-schema.json. *[Revised: 66 was the July count. The test suite has grown since; it passes on the October branch.]*

## Appendix E: Detailed Comparator Diagnostics



**Figure E1.** Detailed comparator diagnosis supporting Figure 2C. Macron-suite alternatives are comparatively similar; the Modi suite contains a low Trump arm on both byte-identical checkpoints. On C, Modi exceeds Trump but is lower than Xi under the one-sided positive-direction test. The figure diagnoses an audit-calibration failure and does not interpret why the base model differs across political figures.